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SSH Log Analysis Lab Report

Using Splunk Basics – SSH Log Analysis Lab Report

Structured Cybersecurity Report

OBJECTIVE

SSH (Secure Shell) is a protocol used for secure remote access to servers. In this lab, we will use Splunk (a powerful log management platform) to analyse SSH authentication logs and spot any unusual login activity. The goals include:

- Uploading Zeke-format SSH logs (JSON) into Splunk.
- Setting up a dedicated index and sourcetype for the SSH data.
- Running searches to examine SSH login attempts and outcomes.
- Detecting failed login attempts and successful connections.
- Identifying potential brute-force or unauthorized access patterns.

Analysing SSH logs is important because repeated login attempts can signal malicious behaviour (for example, an attacker trying to guess a password).

TOOLS AND ENVIRONMENT

This lab was performed using **Splunk**, which is used to search and analyze log data.

The logs used in this analysis are **SSH logs generated by Zeek**. These logs contain information about SSH connections and login attempts.

The data is stored in **JSON format**, which allows Splunk to automatically extract important fields.

A separate index named **ssh_lab** was used to store the logs. This helps in keeping SSH data organized and makes searching easier.

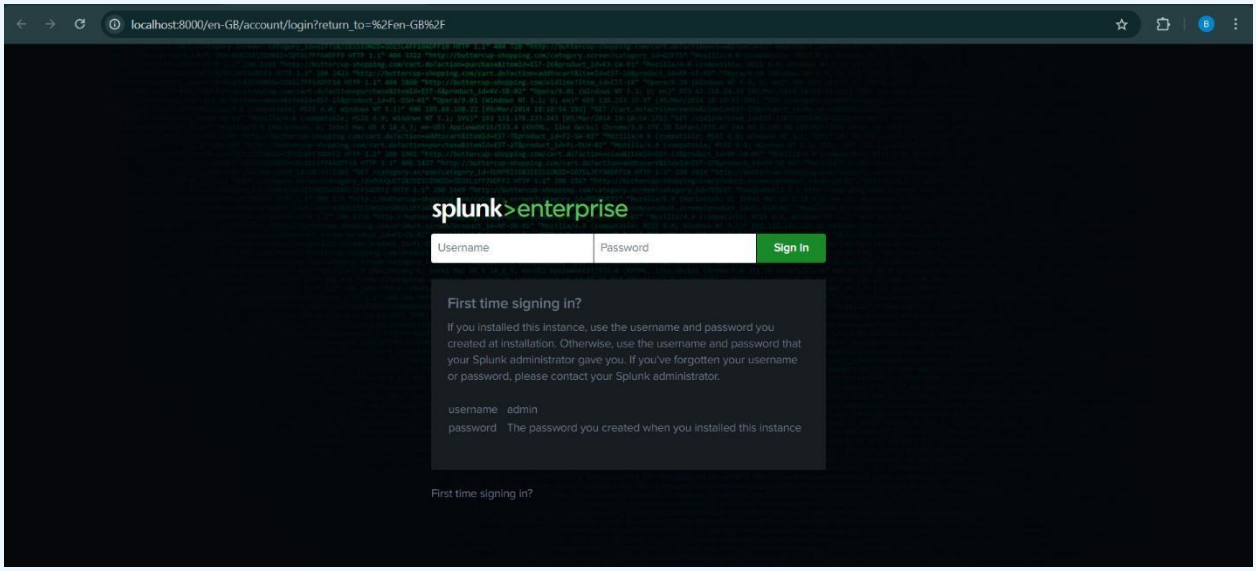
The sourcetype was set to **json**, so that Splunk can correctly read and process the log data.

LAB SETUP

Before performing the analysis, the SSH log file must be uploaded to Splunk.

Step 1 – Open Splunk Web

1. Open **Splunk Web Interface** in your browser.
2. Log in using your credentials.

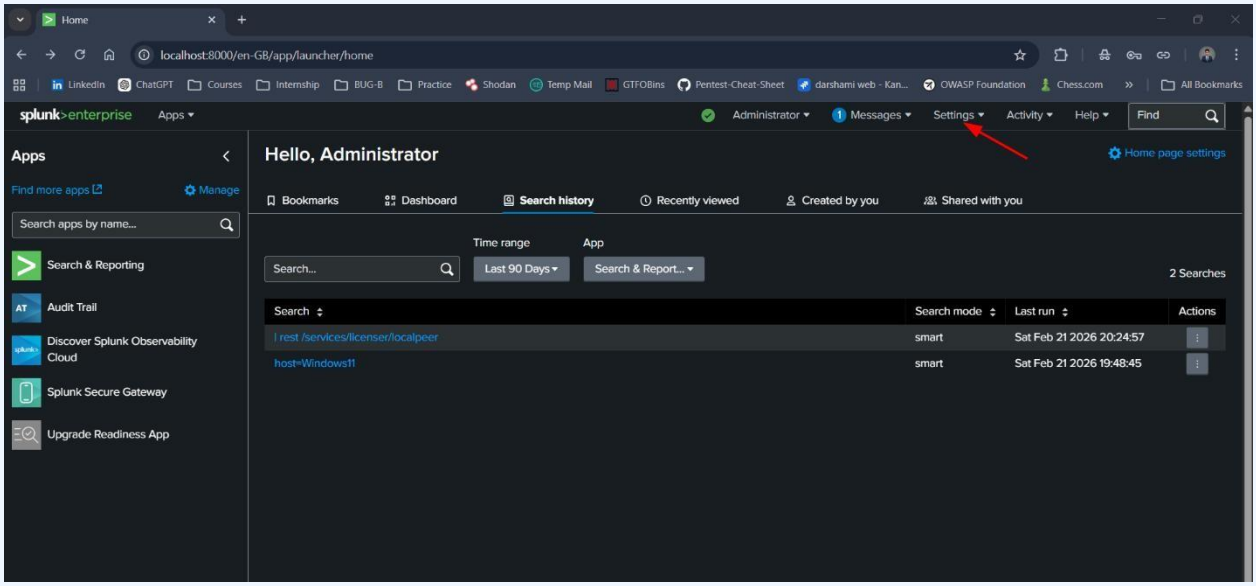


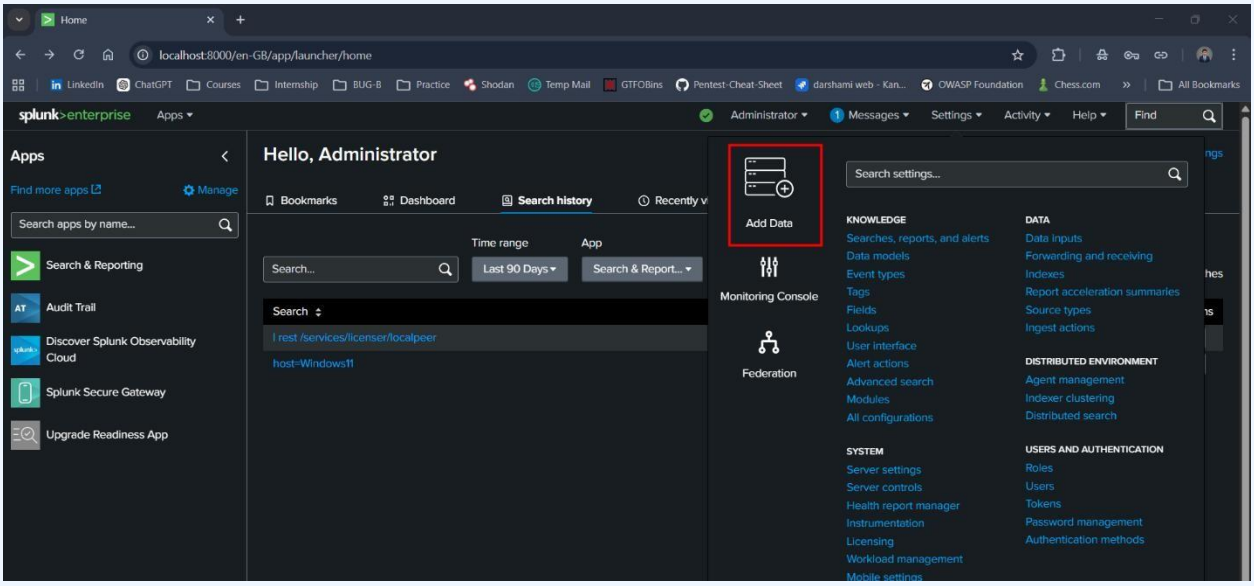
Step 2 – Navigate to Add Data

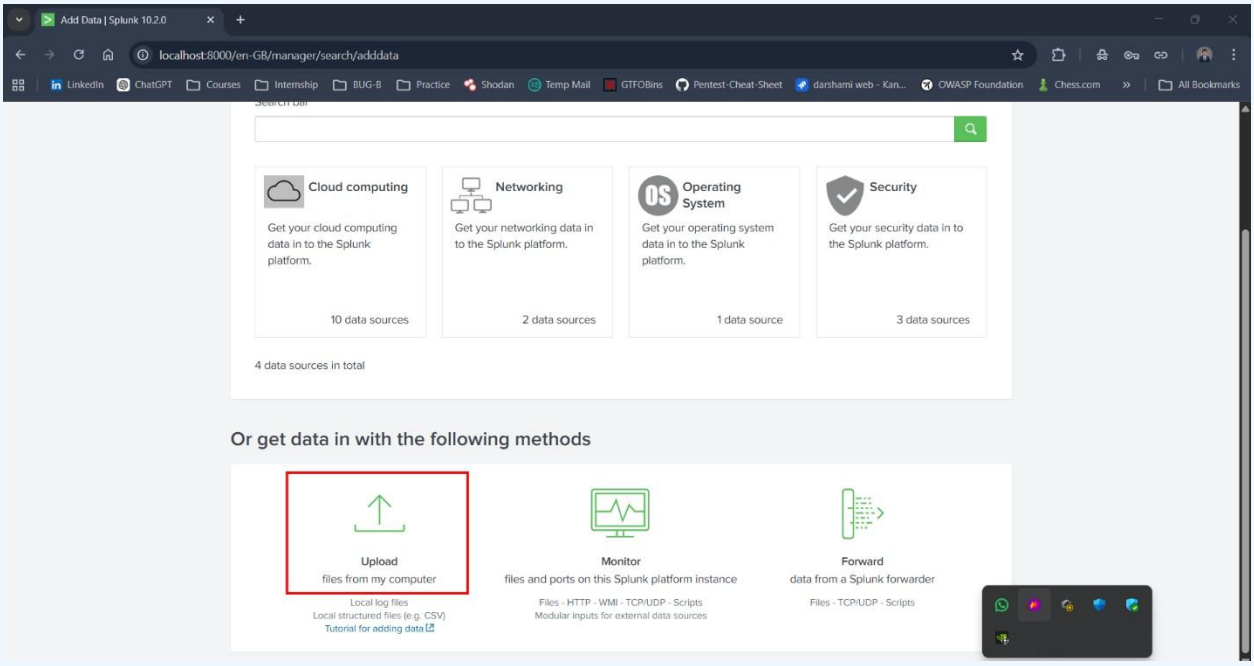
1. Go to:

Settings → Add Data

2. Select **Upload.**







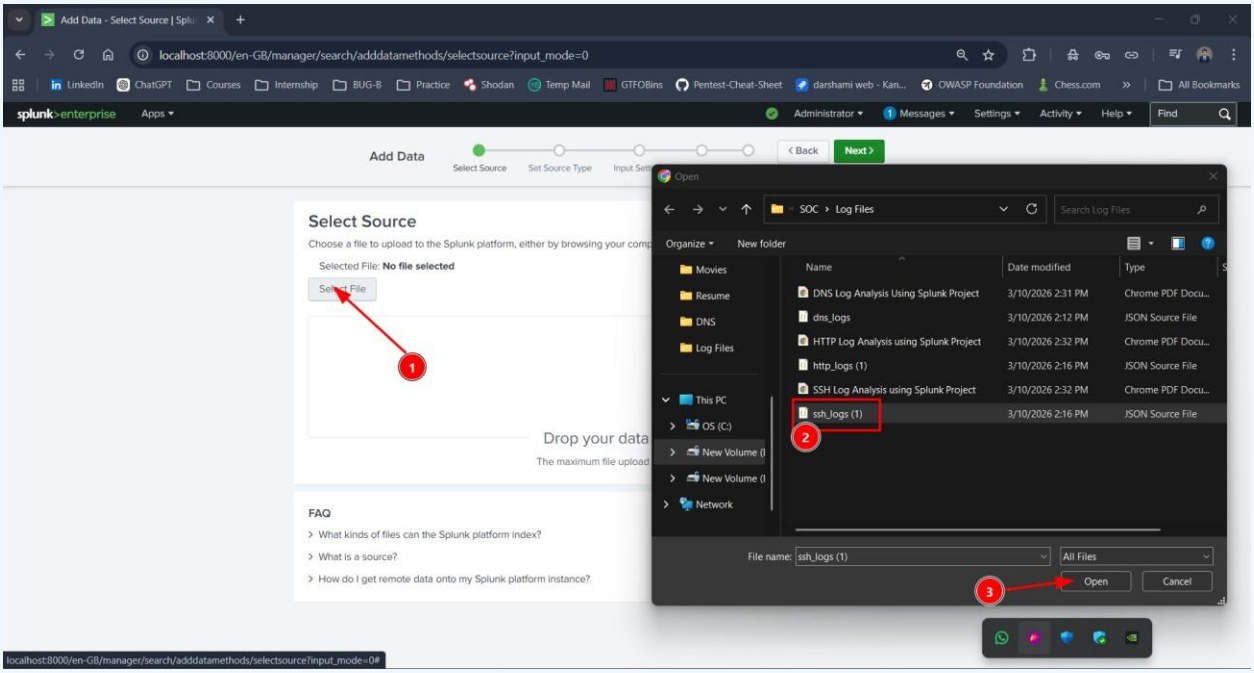
Step 3 – Upload the SSH Log File 1.

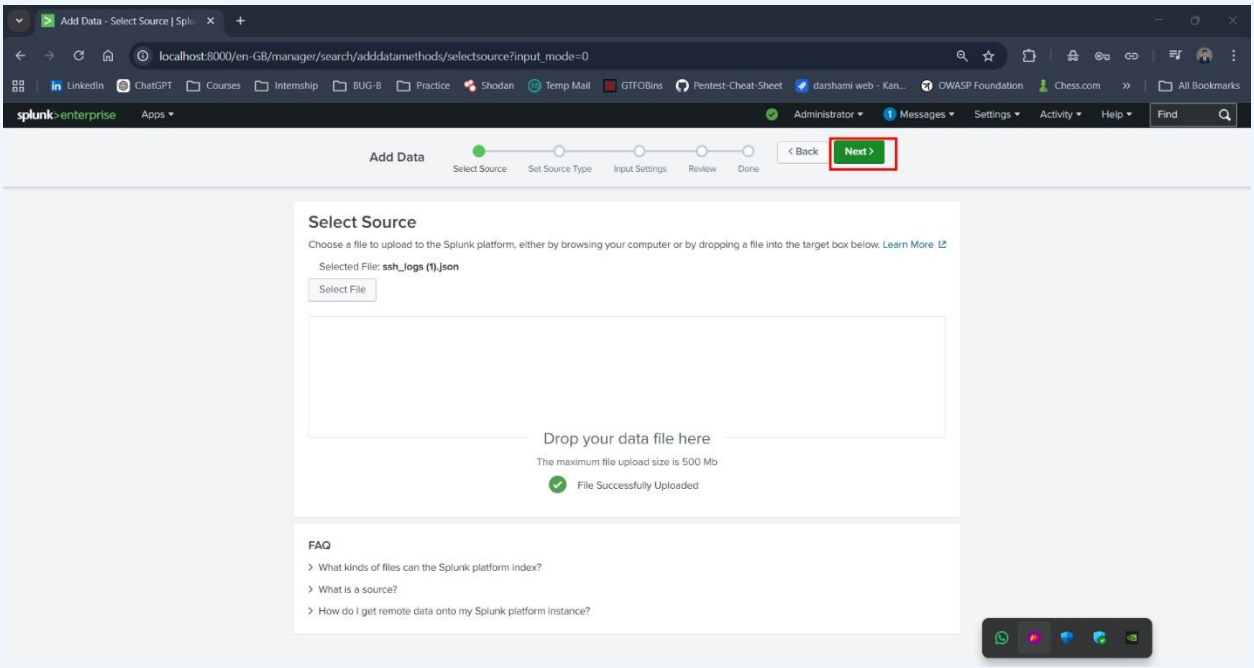
Click **Select File**

2. Upload the file:

ssh_logs

This file contains **SSH connection events captured in Zeek-style format.**





Step 4 – Configure Source Type

Set the source type as:

_json

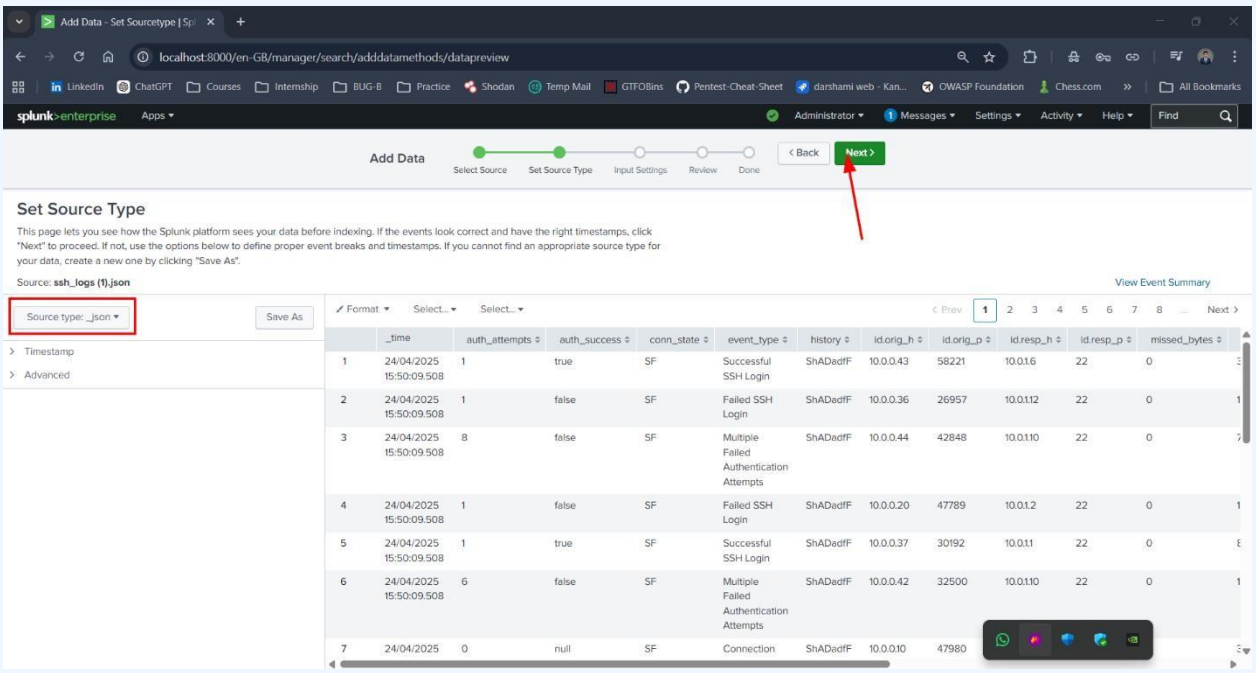
or create a custom sourcetype:

ssh_logs.json

Understanding SSH Logs

SSH logs record each login attempt and connection to a server. Key fields include:

- **id.orig_h:** Source IP address of the client.
- **id.resp_h:** Destination (server) IP address.
- **auth_success:** true if the login succeeded; false if it failed.
- **event_type:** Type of SSH event (e.g., successful, failed, no-auth, multiple-failed). Zeek classifies events into these categories.
- **timestamp:** Date and time of the event.



Step 5 – Configure Index

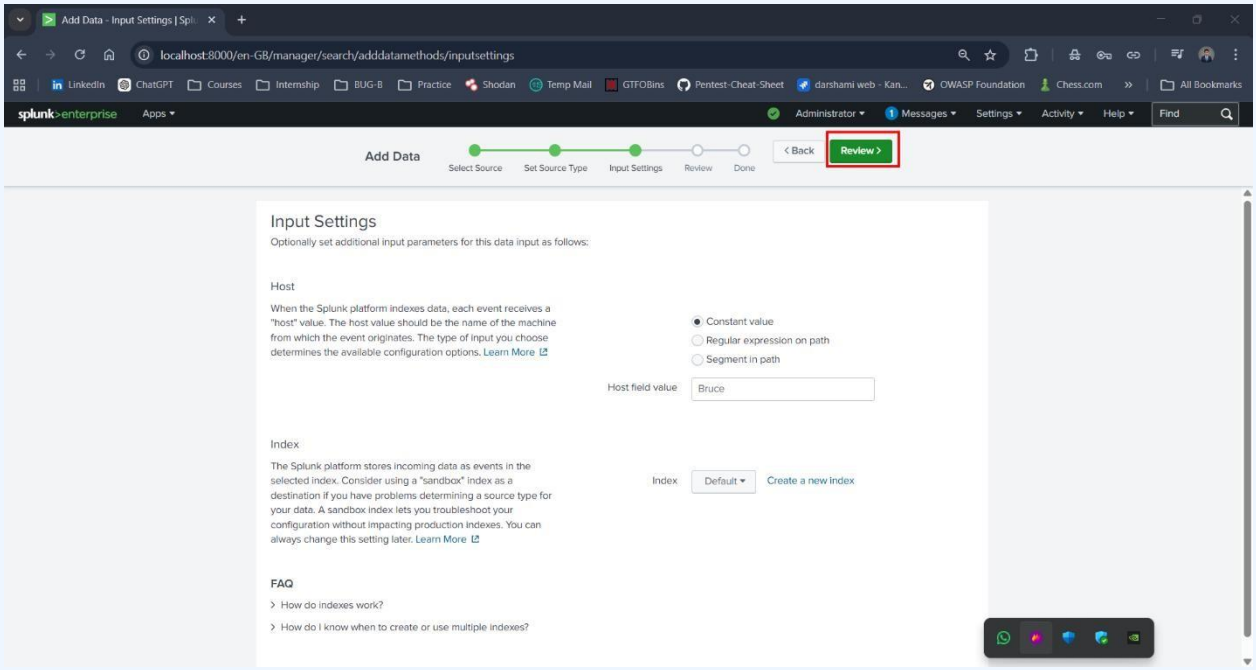
Create or select an index:

ssh_logs

In this step, an index named **ssh_lab** was selected (or created) to store the SSH log data.

Reason

- Keeps SSH-related data separate from other logs
- Helps in organizing the data properly
- Makes searching and analysis easier



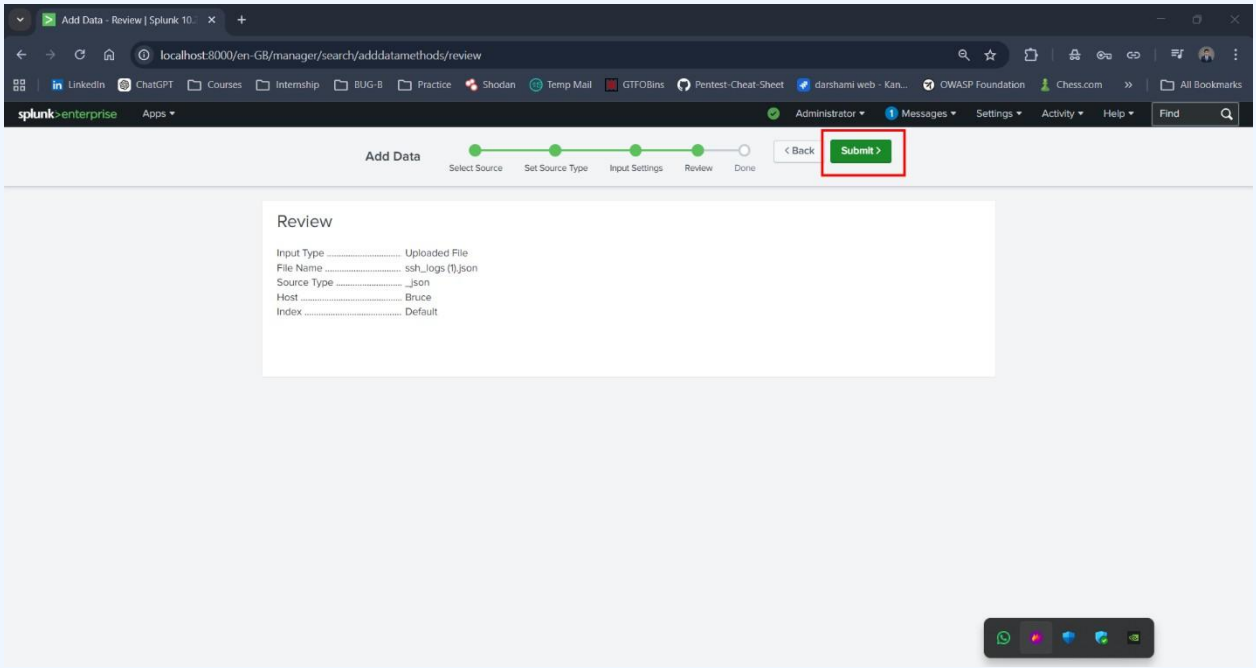
Step 6 – Complete Upload

- Review the configuration settings
- Click on Submit to upload the data
- Splunk will process and index the logs

To verify the upload, run the following search:

```
index=ssh_lab
```

If log events are displayed, the data has been successfully uploaded and indexed.



UNDERSTANDING SSH LOGS

SSH logs contain information about login attempts and connections made to a server using the SSH protocol. These logs are useful for tracking user activity and identifying any unusual or suspicious login behavior.

Each log entry includes details about the source of the connection, the result of authentication, and the type of event that occurred.

Important Fields

The following fields are commonly found in Zeek SSH logs:

- **id.orig_h** – Represents the IP address of the client trying to connect
- **id.resp_h** – Represents the IP address of the destination server
- **auth_success** – Shows whether the login attempt was successful or failed
- **event_type** – Indicates the type of SSH activity (such as successful, failed, etc.)
- **timestamp** – Shows the date and time when the event occurred

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Example SSH Log Event

A typical log entry may look like this:

- Source IP: 192.168.1.10
- Destination IP: 192.168.1.5
- Authentication Status: Failed
- Event Type: failed

This means a client tried to connect to the server but the login attempt was not successful.

```
{
 “id.orig_h”: “192.168.1.10” ,
“ id.resp_h”: “192.168.1.5” ,
 “auth_success” : “false” ,
 “event_type” :  “failed”
 }
```

LAB TASKS

Task 1: List Top 10 Endpoints with Failed SSH Login Attempts

To find which clients are making many failed login attempts, we run:

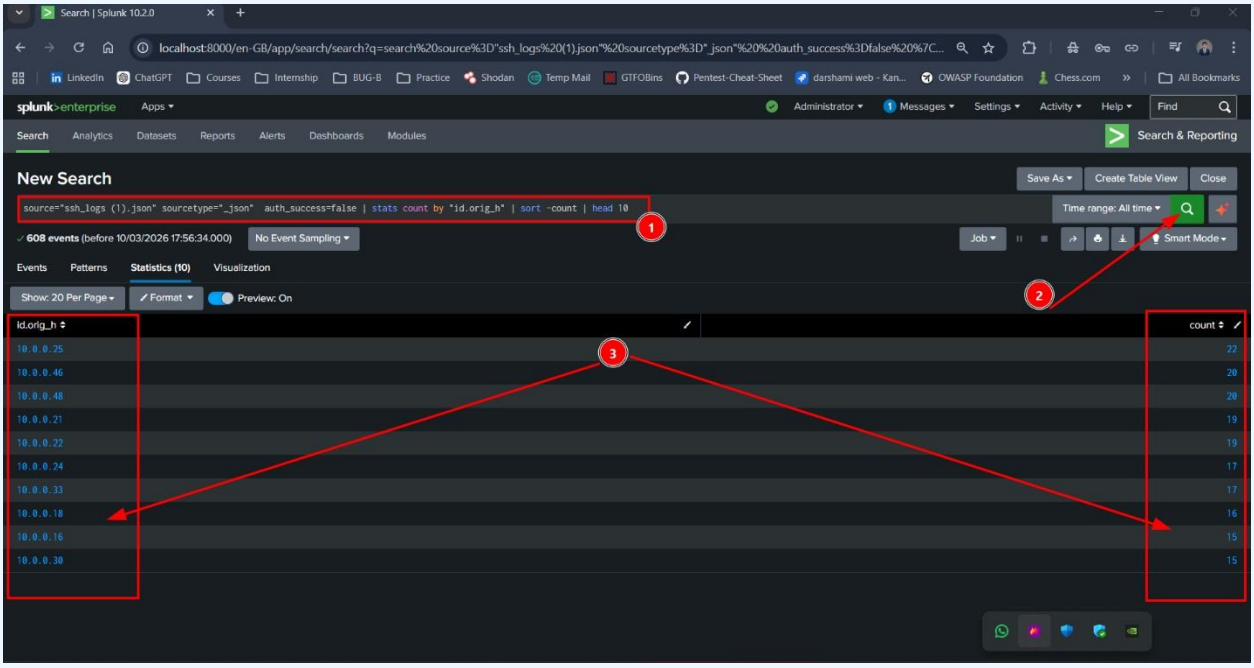
SPL Query

```
index=ssh_lab sourcetype="json" auth_success=false
| stats count by id.orig_h
| sort -count
| head 10
```

Explanation
Steps in this query:

- index=ssh_lab: Search only in our SSH index.
- sourcetype="json": Consider only JSON logs.
- auth_success=false: Filter for failed login events.
- stats count by id.orig_h: Count the number of failures per source IP.
- sort -count: Sort IPs by descending count.
- head 10: Show the top 10 IP addresses.

Analysis:
The result lists the IPs with the most failed logins. For example, if 192.168.1.10 appears with 50 failures, it indicates that address attempted 50 logins and failed each time. This suggests a **brute-force attempt**



Task 2: Count Total SSH Connections

SPL Query

```
index=ssh_lab sourcetype="json"
| stats count as total_ssh_connections
```

Explanation

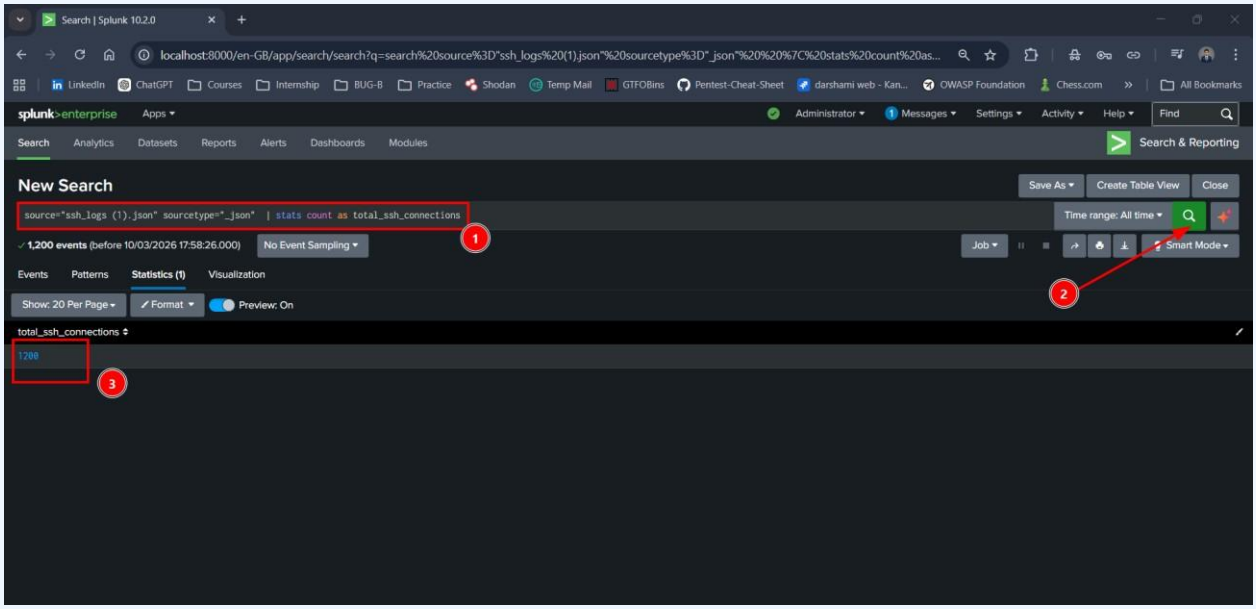
- stats count: Counts all matching events.
- as total_ssh_connections: Renames the output field.

Analysis:

This returns the total number of SSH events (both successful and failed). If it shows 1200, that means there were 1200 login attempts logged. Knowing this total helps us understand normal activity levels.

What This Shows

This query provides the **total number of SSH connection events** recorded in the logs.



This helps understand **overall SSH activity** on the network.

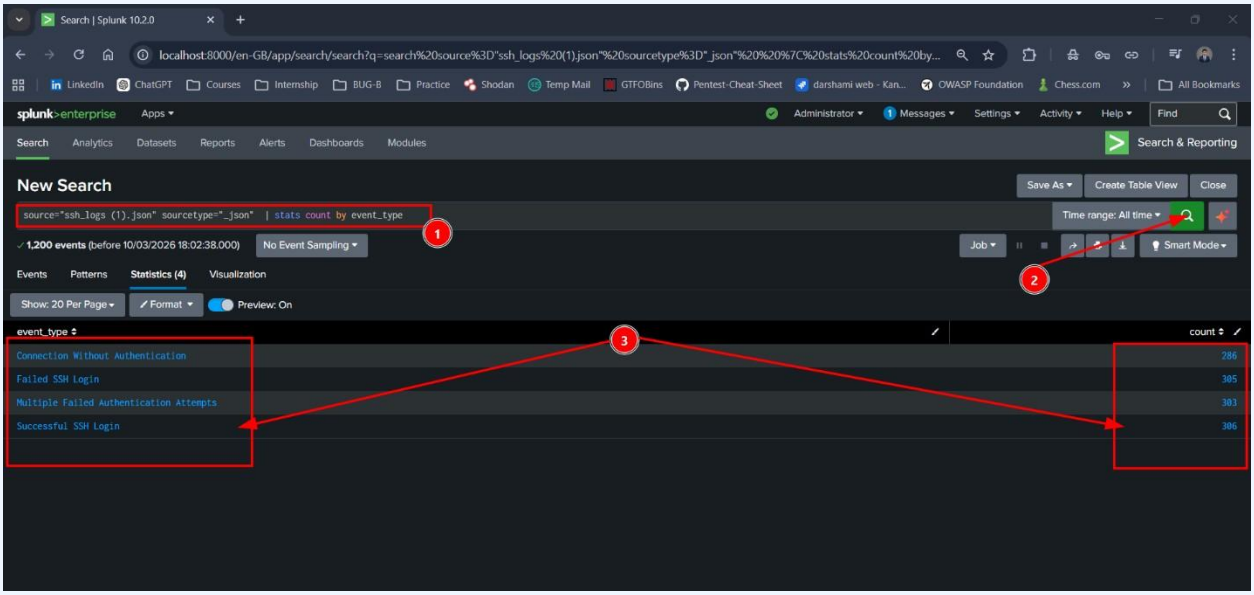
Task 3: Count SSH Event Types

SPL Query

```
index=ssh_lab sourcetype="json"
| stats count by event_type
```

Explanation

- **stats count by event_type:** Counts events for each event_type.



Analysis: Suppose the output is:

- successful: 800
- failed: 350
- no-auth: 40
- multiple-failed: 10

Analyzing SSH logs can help detect:

1. **Brute Force Attacks:** Large numbers of **failed login attempts from one IP**.
2. **Unauthorized Access:** Unexpected **successful logins from unknown IP addresses**.
3. **Suspicious Login Patterns**

Key Observations

Reviewing the SSH log analysis, we can note suspicious patterns:

- **Repeated failures:** An IP with many failed attempts is likely using an automated tool. If one of those attempts succeeds, the attacker has found a valid password.
- **Timing patterns:** If a user usually logs in during the day but a login appears at 2 AM, it's unusual. Unusual login times often hint at unauthorized access.
- **Geographical anomalies:** A login from a foreign country or an unfamiliar IP suggests a compromised credential is being used.
- **Trend alerts:** A sudden increase in "no-auth" events could mean a botnet scanning SSH ports. A surge in "multiple-failed" shows repeated attacks that should be blocked or monitored closely.

Splunk lets us set real-time alerts on these situations. For example, we could alert if more than 10 failures occur in an hour. Dashboards can also visualise these trends, helping security teams respond faster.

CONCLUSION

This lab showed how to use Splunk to analyse SSH logs for security monitoring. We:

- Uploaded and indexed Zeek SSH logs in Splunk.
- Wrote SPL queries to find the top IPs making failed login attempts.
- Counted the total number of SSH connection events.
- Examined the distribution of event types (successful vs failed, etc.).

These steps help uncover security issues:

- **Brute-force attacks:** We can spot IPs with many failed logins.
- **Suspicious activity:** We monitor overall SSH traffic and event trends.
- **Incident readiness:** By understanding normal patterns, we notice when something changes.

Analysing SSH logs is a key part of intrusion detection and system monitoring. Splunk's search and alert features let us detect and respond to threats quickly, keeping the server environment secure.

This type of log analysis is critical for **detecting brute force attacks, monitoring server access, and improving system security**.